

FRONT_BOT

Frontline Reporting & OSINT Node Tracker Bot

Version 0.90

Military conflict	Russian invasion of Ukraine
Opposing parties	Russia – Ukraine
Conflict start date	24. February 2022
Conflict day	1370
Conflict week	196
Reporting period	2025-11-17 → 2025-11-24
GitHub system info	Linux-6.11.0-1018-azure-x86_64-with-glibc2.39 CPU: x86_64 RAM: 8.33 GB
Reporting FRONT_BOT version	0.90
Report writing duration	4 minutes 18 seconds
Project developer	Zsolt Lazar
Developer's website	https://bio.site/zsoltlazar
Project website	https://frontbot.my.canva.site/front-bot

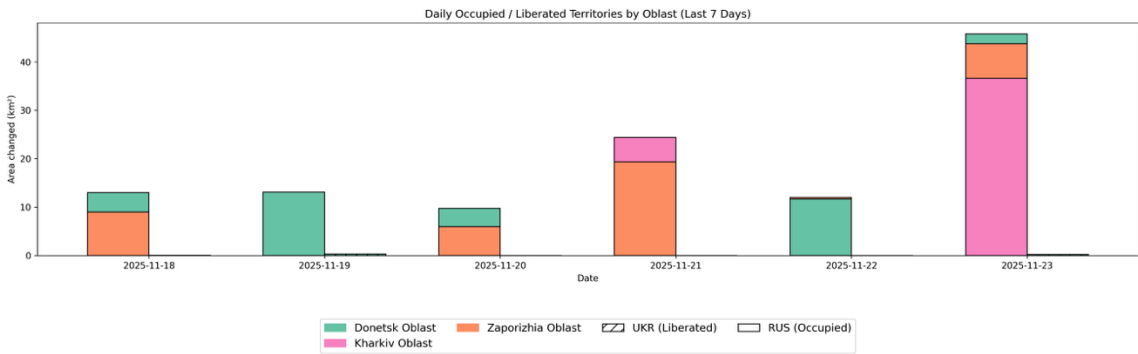
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WEEKLY TERRITORIAL CHANGES

Module 7: Weekly Territorial Change Report (DeepStateMap)

According to DeepStateMap data, between 2025-11-18 and 2025-11-23, Russian and Ukrainian forces continued to contest territory along multiple fronts. Over this 7-day period, Russian gains amounted to 118.0 km², while Ukrainian gains reached 0.6 km². The largest Russian advances were observed in Kharkiv Oblast (41.7 km²), Zaporizhia Oblast (41.7 km²), Donetsk Oblast (34.6 km²). Meanwhile, Ukrainian forces achieved their most notable gains in Donetsk Oblast (0.5 km²), Zaporizhia Oblast (0.0 km²), Kharkiv Oblast (0.0 km²). Overall, the most active oblasts during the week were Zaporizhia Oblast, Kharkiv Oblast, Donetsk Oblast. The peak day for Russian advances was 2025-11-23, while the peak day for Ukrainian advances was 2025-11-19.



Distribution of Gains by Oblast (Last 7 Days)

RUS Gains (118.0 km²)



UKR Gains (0.6 km²)

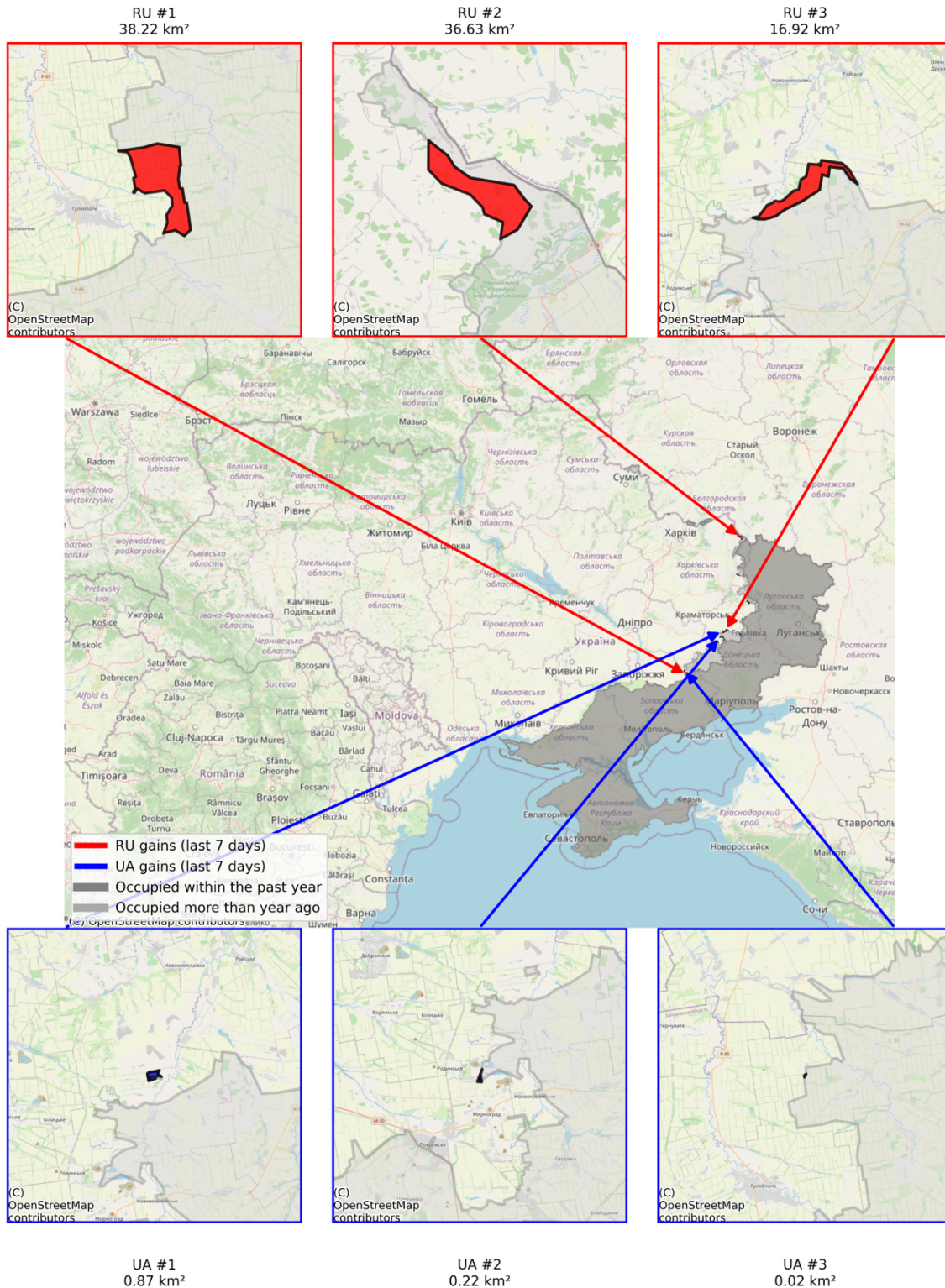


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Module 8: Weekly Territorial Map (DeepStateMap)

Last 7 Days | RU Gains 122.16 km² | UA Gains 1.13 km²
Top 3 territorial changes shown on side maps

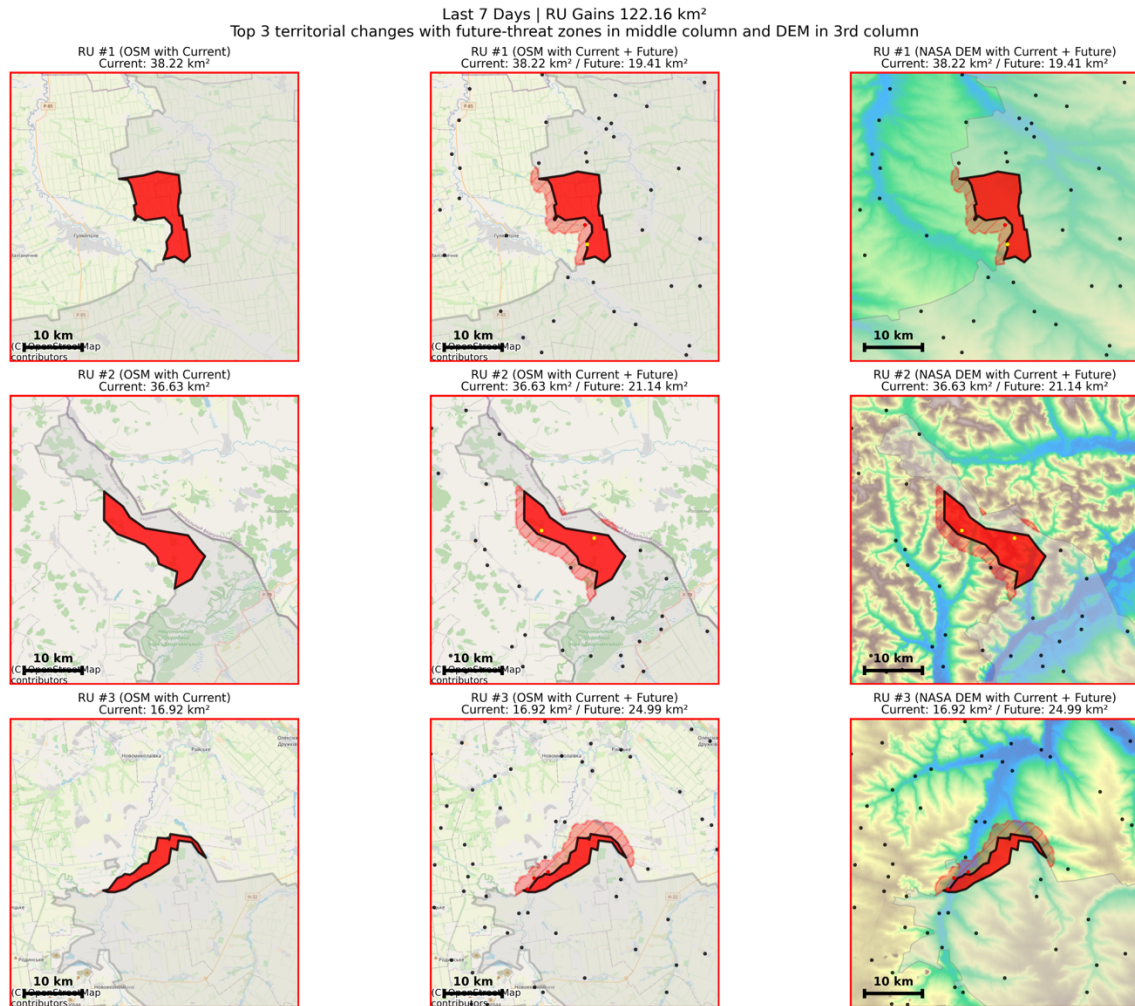


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Module 9: Top 3 Territorial Maps & Narratives (DeepStateMap)



RU Narrative:

According to DeepStateMap data between 2025-11-16 and 2025-11-23, the following territorial developments were observed for RU forces.

7-Day Comparative Summary

Over the past week, RU forces have gained a total of 122.16 km², with the top three territorial gains accounting for 91.77 km² (75% of the total). In comparison, the opposing side has gained 1.13 km² (top three gains: 1.12 km²).

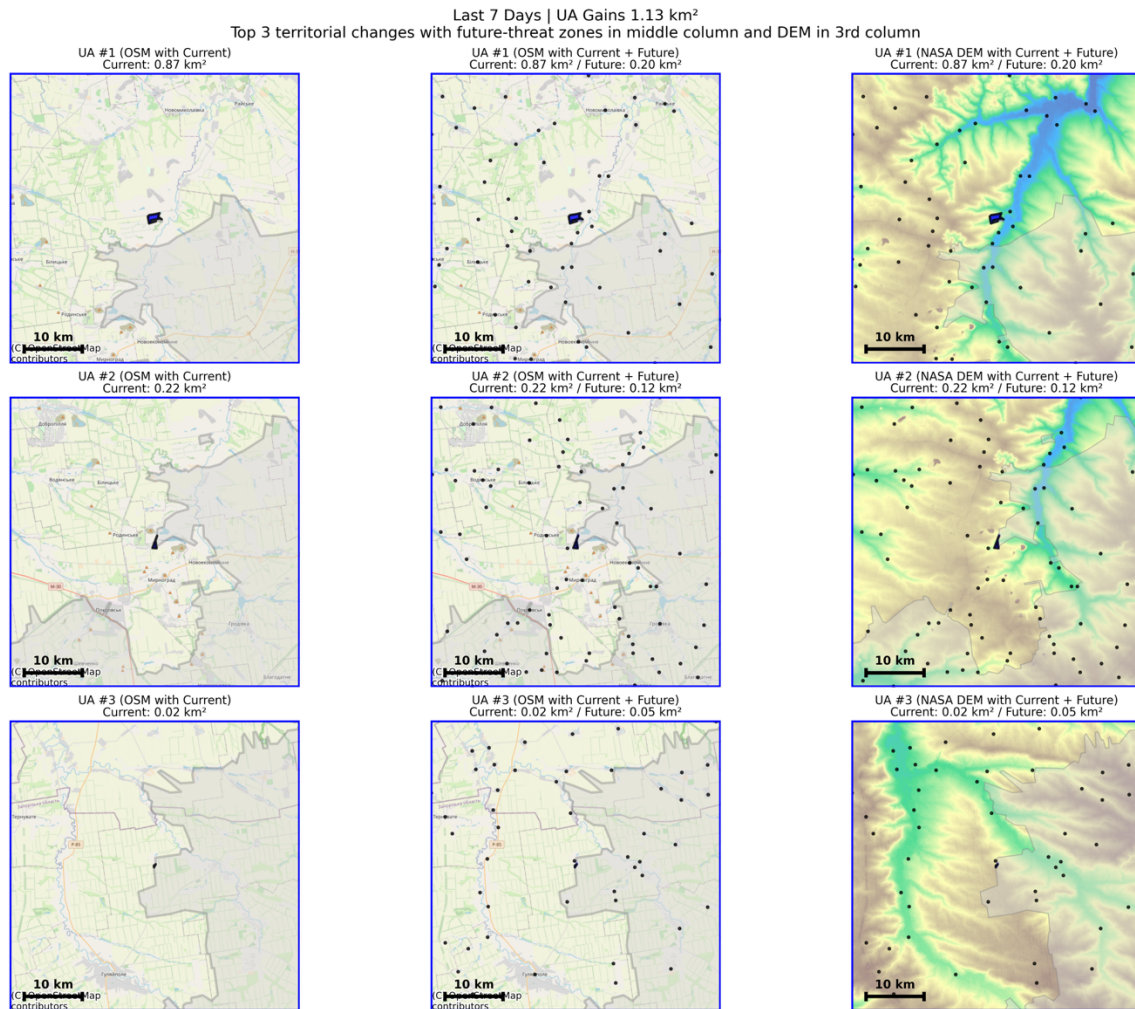
RU Top Territorial Changes Narrative

The #1 largest gain by RU forces covers approximately 38.22 km², which represents about 31% of total gains this week. Currently, the gain affects the settlement Високе. If the advance continues, the following settlements may soon fall within the contested area: Новоуспенівське, Зелений Гай. The #2 largest gain by RU forces covers approximately 36.63 km², which represents about 30% of total gains this week. Currently, the gain affects the settlements: Одрадне, Бологівка. No additional settlements are predicted to be affected in the near future. The #3 largest gain by RU forces covers approximately 16.92 km², which represents about 14%

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of total gains this week. Currently, no settlements are affected by this gain. If the advance continues, the following settlements may soon fall within the contested area: Панківка, Володимирівка, Маяк.



UA Narrative:

According to DeepStateMap data between 2025-11-16 and 2025-11-23, the following territorial developments were observed for UA forces.

7-Day Comparative Summary

Over the past week, UA forces have gained a total of 1.13 km², with the top three territorial gains accounting for 1.12 km² (99% of the total). In comparison, the opposing side has gained 122.16 km² (top three gains: 91.77 km²).

UA Top Territorial Changes Narrative

The #1 largest gain by UA forces covers approximately 0.87 km², which represents about 77% of total gains this week. Currently, no settlements are affected by this gain. No additional

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settlements are predicted to be affected in the near future. The #2 largest gain by UA forces covers approximately 0.22 km², which represents about 20% of total gains this week. Currently, no settlements are affected by this gain. No additional settlements are predicted to be affected in the near future. The #3 largest gain by UA forces covers approximately 0.02 km², which represents about 2% of total gains this week. Currently, no settlements are affected by this gain. No additional settlements are predicted to be affected in the near future.

Module 4: Settlements (DeepStateMap) — Narrative

According to data reported by the DeepStateMap project, between 2025-11-17 to 2025-11-23, the following territorial developments were observed:

Russian forces expanded their control between 2025-11-17 to 2025-11-23, with Russia gaining 280.9 km² (22.9% less than the previous week, which was 364.3 km²), and Ukraine gaining 1.8 km² (9.2% more than the previous week, which was 1.7 km²).

The most occupied oblast was Donetsk Oblast (Новоспаське, Покровськ, Мироліувка). Newly occupied settlements include: Бологівка (village, Kharkiv Oblast), Високе (village, Zaporizhia Oblast), Мироліувка (village, Donetsk Oblast), Новоспаське (town, Donetsk Oblast), Одрадне (village, Kharkiv Oblast), Покровськ (town, Donetsk Oblast).

Based on the local weekly advance (median ≈ 0.81 km, dynamic buffer 814 m), settlements likely next affected by Russian advances include Щербинівка (town, Donetsk Oblast) (45 m), Маяк (village, Donetsk Oblast) (56 m), Михайлівка (village, Donetsk Oblast) (135 m), Борівська Андріївка (village, Kharkiv Oblast) (138 m), Гнатівка (village, Donetsk Oblast) (314 m), Володимирівка (village, Donetsk Oblast) (410 m), Разіне (village, Donetsk Oblast) (428 m), Панківка (village, Donetsk Oblast) (477 m), Pir (village, Donetsk Oblast) (806 m).

Module 1: Equipment Losses Weekly Comparison (WarSpotting)— Narrative

According to data reported by the WarSpotting project, between 2025-11-16 and 2025-11-22, Russian forces lost 16 pieces of equipment (cumulative total: 21875), representing a decrease of 61.9% compared to the previous week.

The largest increase was seen in Transport, which rose to 11 units (+1, +10.0%). Meanwhile, Infantry fighting vehicles losses dropped to 3 units (-14, -82.4%).



Weekly Retrospective (last 52 weeks):

- Average weekly losses over the past year: 111.6, current week below average (16 units).
- Largest weekly spike: 742 units in week ending 2022-03-28.
- Cumulative total: 21875

Equipment categories with the highest relative increase compared to their yearly average:

- Other (+92.6%)
- Transport (+4.8%)

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- Infantry fighting vehicles (+90.0%)

Categories with the largest decreases:

- Airplanes (-100.0%)
- Ambulances, medical vehicles (-100.0%)
- Anti-aircraft systems (-100.0%)

Geolocation data was available for 7 of these losses (43.8%), compared to 31 (73.8%) the week before.

The most frequently reported location was Malynivka, Pokrovsk raion, indicating sustained or intensified activity in the area.

Module 2: Equipment Losses Weekly Comparison (Oryx)— Narrative with Retrospective & Category Context

According to data reported by the Oryx project, between 2025-11-17 and 2025-11-23, Russian forces lost 22 units (cumulative total: 23413), while Ukrainian forces lost 38 units (cumulative total: 10523). This represents a increase of 0.1% for Russia and a increase of 0.4% for Ukraine compared to the previous week. The RU/UA loss ratio was 2.22, compared to 2.23 the previous week.



Weekly Retrospective (last 52 weeks):

- Russia: average weekly losses over the past year 11.6, current week above average (22 units), largest weekly spike: 252 units in week ending 2025-10-24.
- Ukraine: average weekly losses over the past year 9.4, current week above average (38 units), largest weekly spike: 236 units in week ending 2025-03-20.
- Cumulative totals: Russia 23413, Ukraine 10523

◆ Per-Category Weekly Context and Top 3 Highlights:

Russia:

- Tanks: 2 (above average 1.8)
- AFV: -4 (below average 1.6)
- IFV: 8 (above average 3.4)
- APC: -1 (below average 0.4)
- IMV: 1 (above average 0.3)
- Engineering: 1 (above average 0.3)
- Coms: 1 (above average 0.0)
- Vehicles: 13 (above average 1.6)
- Aircraft: 0 (below average 0.2)
- Infantry: 8 (above average 4.1)
- Logistics: 2 (above average 0.3)
- Armor: -2 (below average 3.3)
- Anti-air: 0 (below average 0.3)
- Artillery: 0 (below average 0.3)

▲ Top 3 increases this week: Vehicles (+11.4), IFV (+4.6), Infantry (+3.9)

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▼ Top 3 decreases this week: APC (-1.4), Armor (-5.3), AFV (-5.6)

Ukraine:

- Tanks: 2.0 (above average 1.0)
- AFV: 1.0 (above average 0.2)
- IFV: 1.0 (below average 1.1)
- APC: 9.0 (above average 1.5)
- IMV: 6.0 (above average 1.8)
- Engineering: 0.0 (below average 0.2)
- Coms: -1.0 (below average 0.1)
- Vehicles: 2.0 (above average 0.7)
- Aircraft: 0.0 (below average 0.0)
- Infantry: 16.0 (above average 4.4)
- Logistics: -1.0 (below average 0.3)
- Armor: 3.0 (above average 1.2)
- Anti-air: 0.0 (below average 0.1)
- Artillery: 2.0 (above average 0.8)

▲ Top 3 increases this week: Infantry (+11.6), APC (+7.5), IMV (+4.2)

▼ Top 3 decreases this week: Engineering (-0.2), Coms (-1.1), Logistics (-1.3)

Russia units with the highest relative increase compared to their weekly average were Vehicles (+11.4), IFV (+4.6), Infantry (+3.9). Units showing the largest decreases were APC (-1.4), Armor (-5.3), AFV (-5.6). This indicates which equipment types were most active or affected on the frontlines this week.

Ukraine units with the highest relative increase compared to their weekly average were Infantry (+11.6), APC (+7.5), IMV (+4.2). Units showing the largest decreases were Engineering (-0.2), Coms (-1.1), Logistics (-1.3). This indicates which equipment types were most active or affected on the frontlines this week.

Module 3: Occupation Change Analysis (DeepStateMap) — Narrative

According to data reported by the DeepStateMap project, as of 2025-11-23, Russian forces occupy approximately 95889.02 km² in Ukraine, which is 15.95% of the country's total territory. In the last 7 days, the occupied territory has increased by 121.03 km², a growth of 0.02%. Luhansk Oblast has 26548.29 km² occupied, accounting for 99.52% of its territory, which changed by 0.00 km² (0.00%) over the past week. Autonomous Republic of Crimea has 26846.21 km² occupied, accounting for 98.96% of its territory, which changed by 0.00 km² (0.00%) over the past week. Donetsk Oblast has 20362.49 km² occupied, accounting for 76.29% of its territory, which changed by 37.63 km² (0.14%) over the past week.

Module 4: Settlements (DeepStateMap) — Narrative - Future

Settlements likely to be next affected based on current frontline dynamics:

- Potentially threatened by Russian advance (avg advance ≈ 0.81 km, buffer 814 m):
 - Щербинівка (town, Donetsk Oblast) (45 m)

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- Маяк (village, Donetsk Oblast) (56 m)
- Михайлівка (village, Donetsk Oblast) (135 m)
- Борівська Андріївка (village, Kharkiv Oblast) (138 m)
- Гнатівка (village, Donetsk Oblast) (314 m)
- Володимирівка (village, Donetsk Oblast) (410 m)
- Разіне (village, Donetsk Oblast) (428 m)
- Панківка (village, Donetsk Oblast) (477 m)
- Pir (village, Donetsk Oblast) (806 m)

Module 5: Settlements (DeepStateMap) — Narrative

This briefing focuses on Ukraine overall and on oblasts where occupation levels are between 2% and 95%. Oblasts with less than 2% occupation or more than 95% occupation are excluded from this analysis.

As of the latest update, Ukraine overall is 15.9% occupied. At the current weekly pace, full occupation of Ukraine would take approximately 4174.7 weeks, or about 80.3 years.

Key oblasts under partial occupation:

Donetsk Oblast is currently 76.3% occupied. Over the past week, the occupied area changed by 37.6 km² (0.14%). current trends suggest full occupation could occur in approximately 168.2 weeks (~3.2 years). Zaporizhia Oblast is currently 74.3% occupied. Over the past week, the occupied area changed by 41.7 km² (0.15%). current trends suggest full occupation could occur in approximately 166.8 weeks (~3.2 years). Kharkiv Oblast is currently 4.7% occupied. Over the past week, the occupied area changed by 41.7 km² (0.13%). current trends suggest full occupation could occur in approximately 724.2 weeks (~13.9 years).

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Module 10: Settlement level - Покровськ (DeepStateMap)



Summary Report (Покровськ)

According to DeepStateMap data between 2025-11-18 and 2025-11-24, the following territorial developments were observed within the settlement of Покровськ.

The settlement covers a total area of approximately 31.17 km². As of the latest reporting date (2025-11-24), around 13.06 km² of its territory is under occupation, representing roughly 41.9% of the entire settlement. The frontline first entered Покровськ on 2025-09-29. Over the past seven days, territorial gains within the settlement amounted to 3.57 km².

The largest single-day advance occurred on 2025-11-22, with approximately 2.93 km² gained. The smallest daily gain was recorded on 2025-11-18, with an increase of 0.00 km².

Daily gains inside settlement (Покровськ)

2025-11-18 — Gain inside settlement: 0.000 km²
2025-11-19 — Gain inside settlement: 0.000 km²
2025-11-20 — Gain inside settlement: 0.000 km²
2025-11-21 — Gain inside settlement: 0.000 km²
2025-11-22 — Gain inside settlement: 2.925 km²

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2025-11-23 — Gain inside settlement: 0.000 km²

2025-11-24 — Gain inside settlement: 0.641 km²

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RAW DATA REPORTS

Module 1: Equipment Losses Weekly Comparison (WarSpotting)— RAW

 Equipment Losses Report (2025-11-16 to 2025-11-22):

- Total reported equipment losses: 16 ( 26 from previous week, -61.9%)

◆ Breakdown by equipment type:

- Anti-aircraft systems: 0 ( -3, -100.0%)
- Command posts, communication: 0 ( -1, -100.0%)
- Engineering: 0 ( -1, -100.0%)
- Infantry fighting vehicles: 3 ( -14, -82.4%)
- Infantry mobility vehicles: 0 ( -1, -100.0%)
- Other: 1 ( +0, +0.0%)
- Tanks: 1 ( -7, -87.5%)
- Transport: 11 ( +1, +10.0%)

 Geolocation Summary:

- Last week: 7 geolocated out of 16 (43.8%)
- Previous week: 31 geolocated out of 42 (73.8%)

 Units Most Affected (Last Week):

 Locations with Most Losses (Last Week):

- Malynivka, Pokrovsk raion: 3 losses
- Mala Tokmachka, Polohy raion: 3 losses
- Donetsk oblast: 2 losses

Module 2: Equipment Losses Weekly Comparison (Oryx)— RAW

 Weekly Equipment Losses Report (2025-11-17 to 2025-11-23):

 Russia (cumulative): 23413 (Δ +22, +0.1%)

 Ukraine (cumulative): 10523 (Δ +38, +0.4%)

 RU/UA Ratio: 2.22 (prev 2.23)

◆ Breakdown by equipment type (Russia, weekly increments):

- Tanks: 2 ( -6, -75.0%)
- AFV: -4 ( -9, -180.0%)
- IFV: 8 ( -7, -46.7%)
- APC: -1 ( -2, -200.0%)
- IMV: 1 ( +1, +0.0%)

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- Engineering: 1 (📈 +1, +0.0%)
- Coms: 1 (📈 +1, +0.0%)
- Vehicles: 13 (📈 +2, +18.2%)
- Aircraft: 0 (📉 -1, -100.0%)
- Infantry: 8 (📉 -8, -50.0%)
- Logistics: 2 (📈 +2, +0.0%)
- Armor: -2 (📉 -15, -115.4%)
- Antiair: 0 (📉 -4, -100.0%)
- Artillery: 0 (📊 +0, +0.0%)

◆ Breakdown by equipment type (Ukraine, weekly increments):

- Tanks: 2.0 (📉 -9.0, -81.8%)
- AFV: 1.0 (📊 +0.0, +0.0%)
- IFV: 1.0 (📉 -8.0, -88.9%)
- APC: 9.0 (📉 -14.0, -60.9%)
- IMV: 6.0 (📉 -39.0, -86.7%)
- Engineering: 0.0 (📉 -1.0, -100.0%)
- Coms: -1.0 (📉 -4.0, -133.3%)
- Vehicles: 2.0 (📉 -7.0, -77.8%)
- Aircraft: 0.0 (📊 +0.0, +0.0%)
- Infantry: 16.0 (📉 -61.0, -79.2%)
- Logistics: -1.0 (📉 -5.0, -125.0%)
- Armor: 3.0 (📉 -9.0, -75.0%)
- Antiair: 0.0 (📉 -2.0, -100.0%)
- Artillery: 2.0 (📉 -17.0, -89.5%)

Module 3: Occupation Change Analysis (DeepStateMap) — Detailed

Occupation report for Ukraine comparing 2025-11-16 to 2025-11-23:

Overall, occupied territory has increased by 121.03 km² (0.02 percentage points).

Top 3 oblasts with increased occupation:

- Kharkiv Oblast: +41.72 km² (0.13%)
- Zaporizhia Oblast: +41.67 km² (0.15%)
- Donetsk Oblast: +37.63 km² (0.14%)

Top 3 oblasts with decreased occupation:

- Luhansk Oblast: 0.00 km² (0.00%)
- Autonomous Republic of Crimea: 0.00 km² (0.00%)
- Sumy Oblast: 0.00 km² (0.00%)

Oblasts with significant occupation (>10% occupied):

- Luhansk Oblast: 99.52%
- Autonomous Republic of Crimea: 98.96%
- Donetsk Oblast: 76.29%
- Zaporizhia Oblast: 74.34%

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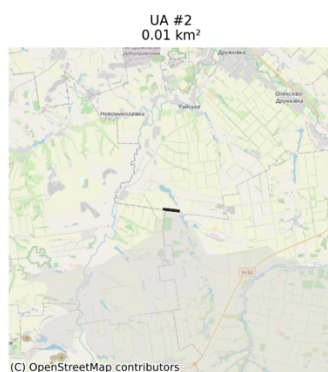
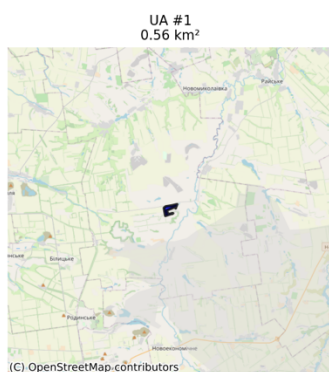
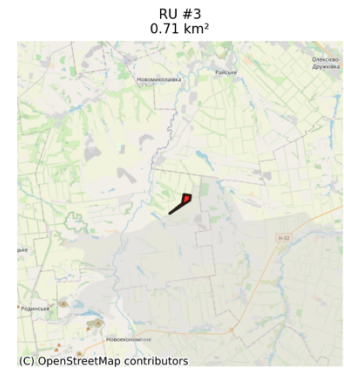
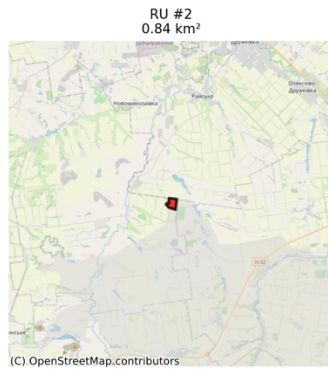
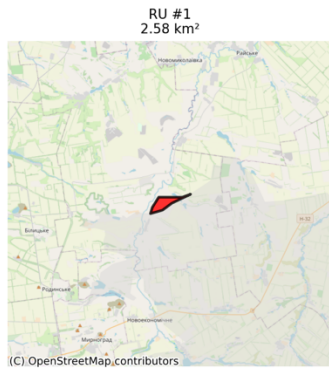
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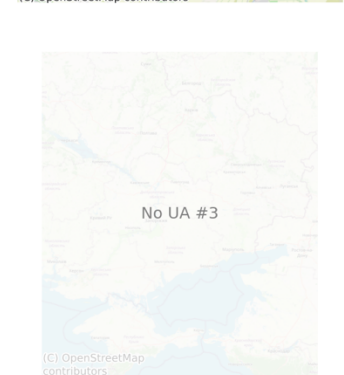
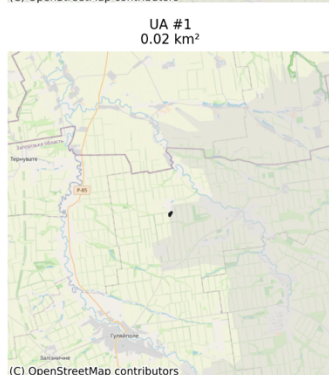
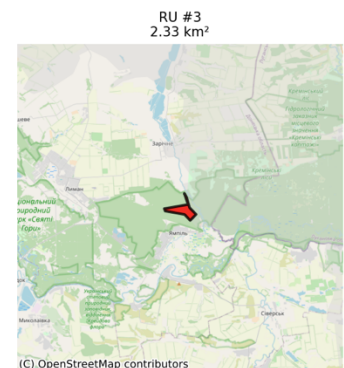
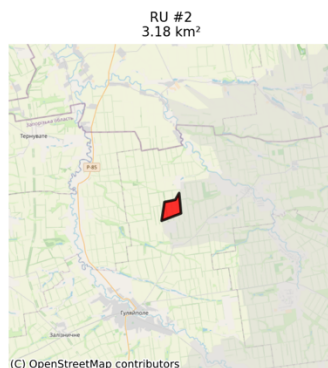
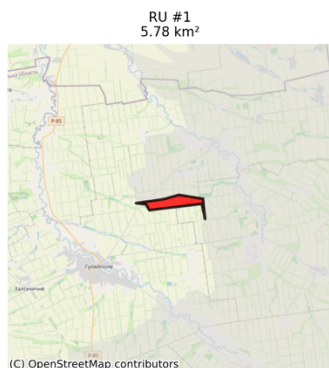
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Module 6: Frontline Map Report

2025-11-17 | Daily Total: RU: 4.13 km² | UA: 0.58 km² (Top 3 gains shown)



2025-11-18 | Daily Total: RU: 13.02 km² | UA: 0.03 km² (Top 3 gains shown)



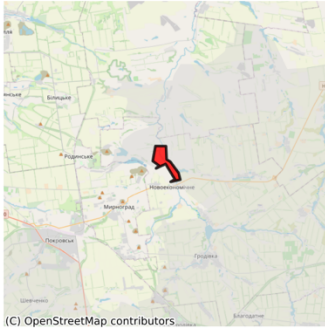
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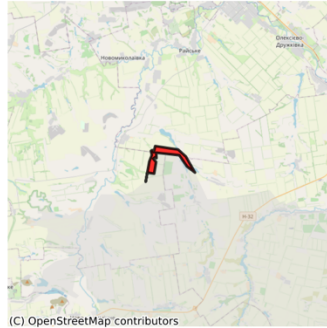
2025-11-19 | Daily Total: RU: 13.09 km² | UA: 0.31 km² (Top 3 gains shown)

RU #1
3.76 km²



(C) OpenStreetMap contributors

RU #2
3.75 km²



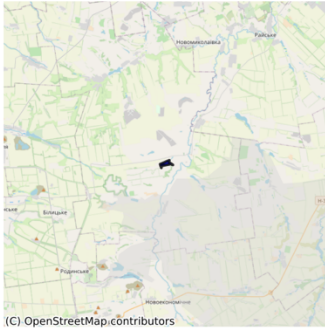
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RU #3
3.75 km²



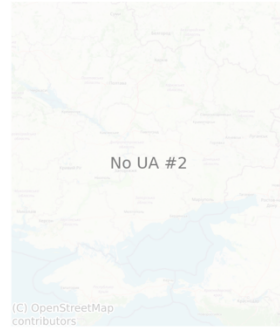
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UA #1
0.31 km²



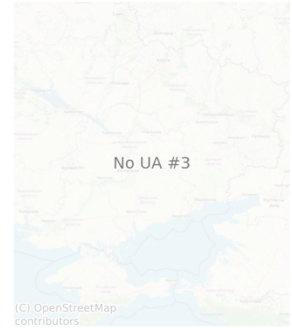
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No UA #2



(C) OpenStreetMap contributors

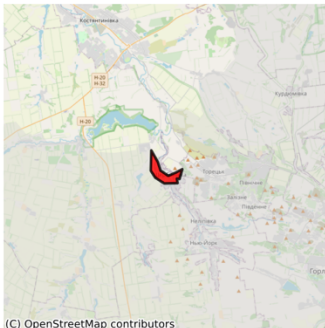
No UA #3



(C) OpenStreetMap contributors

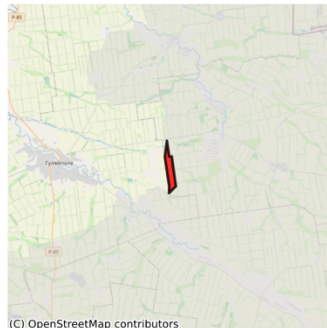
2025-11-20 | Daily Total: RU: 9.74 km² | UA: 0.00 km² (Top 3 gains shown)

RU #1
3.78 km²



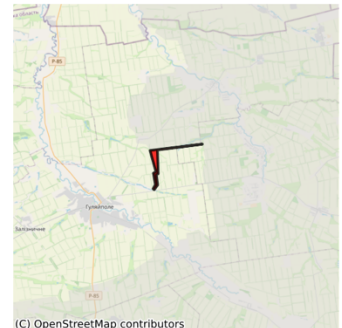
(C) OpenStreetMap contributors

RU #2
2.78 km²



(C) OpenStreetMap contributors

RU #3
1.74 km²



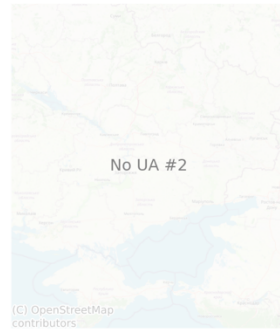
(C) OpenStreetMap contributors

UA #1
0.00 km²



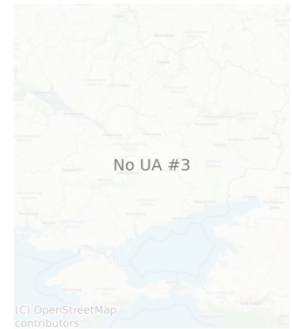
(C) OpenStreetMap contributors

No UA #2



(C) OpenStreetMap contributors

No UA #3



(C) OpenStreetMap contributors

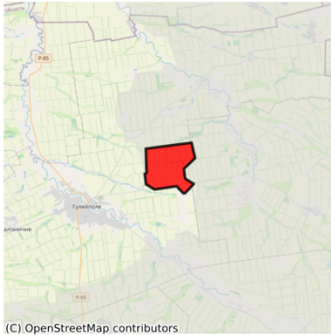
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2025-11-21 | Daily Total: RU: 24.43 km² | UA: 0.00 km² (Top 3 gains shown)

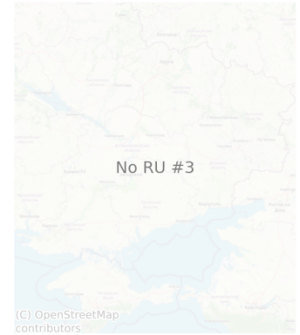
RU #1
19.34 km²



RU #2
5.09 km²



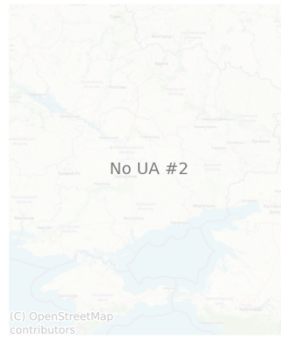
No RU #3



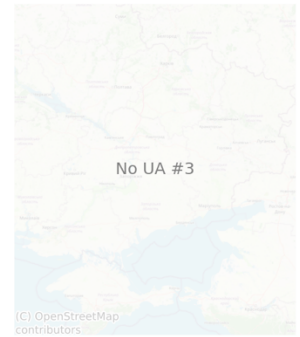
UA #1
0.00 km²



No UA #2

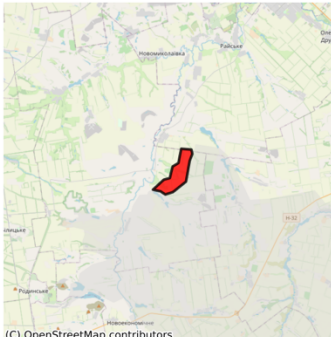


No UA #3

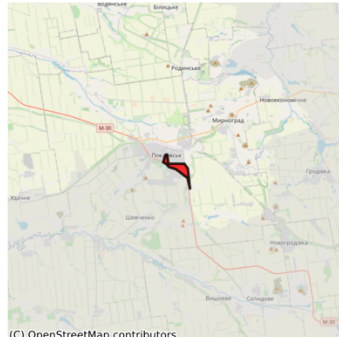


2025-11-22 | Daily Total: RU: 11.97 km² | UA: 0.00 km² (Top 3 gains shown)

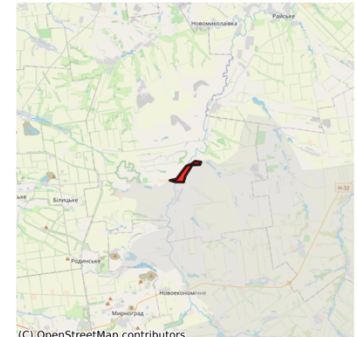
RU #1
6.90 km²



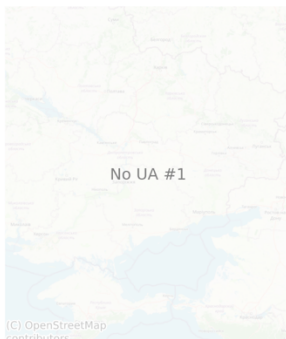
RU #2
1.85 km²



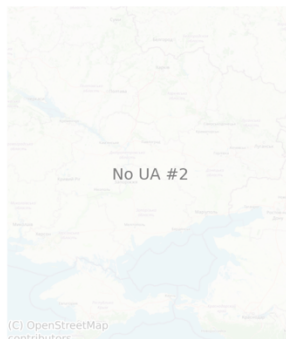
RU #3
1.49 km²



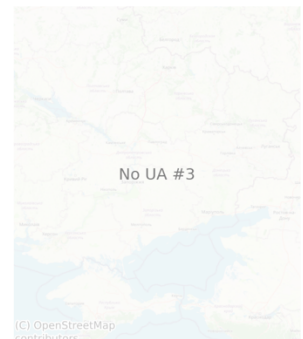
No UA #1



No UA #2



No UA #3



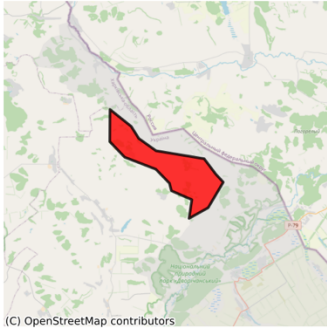
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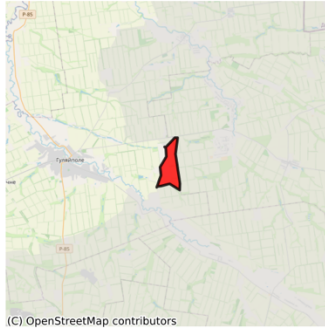
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2025-11-23 | Daily Total: RU: 45.81 km² | UA: 0.22 km² (Top 3 gains shown)

RU #1
36.63 km²



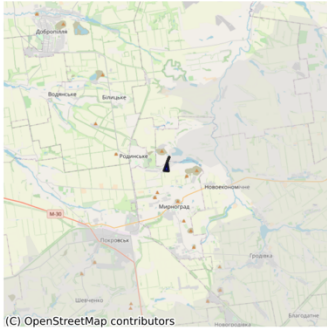
RU #2
7.14 km²



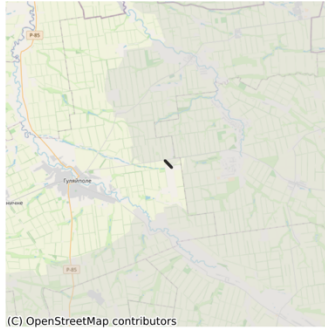
RU #3
2.03 km²



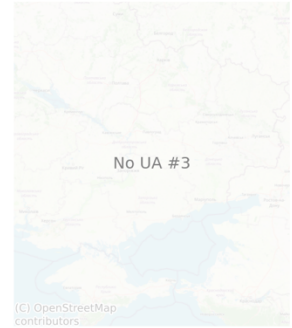
UA #1
0.22 km²



UA #2
0.00 km²



No UA #3



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METHODOLOGY

Module 1: Equipment Losses Weekly Comparison (WarSpotting) relies on open-source intelligence from the WarSpotting database, which aggregates visually confirmed Russian equipment losses reported during the conflict in Ukraine. The module uses an automated pipeline to download the most recent dataset from a controlled Kaggle repository if it is not already present locally. After ingestion, the data is cleaned by parsing the date field, removing invalid entries, and retaining key attributes such as equipment type, unit affiliation, nearest location, and geocoordinates where available. The analytical approach is based on a comparative weekly framework: the last seven-day period ending on the most recent entry is contrasted with the preceding week. For both timeframes, the module calculates total losses, the distribution of equipment types, the proportion of geolocated cases, and the most frequently affected units and locations. It then determines absolute and percentage changes in losses across weeks, identifying the largest increases or decreases by equipment category. Outputs are generated in two forms: a structured statistical summary presenting detailed counts and trends, and a natural-language narrative that highlights key findings in descriptive form.

Module 2: Equipment Losses Weekly Comparison (Oryx) performs a similar comparative analysis of equipment losses but uses the Oryx dataset, which is based on visually verified equipment losses for both Russian and Ukrainian forces. The data is sourced from an automated Kaggle repository and is structured as daily cumulative counts across multiple equipment categories for each side. After loading, dates are standardized and all numerical fields are converted to consistent numeric formats to ensure accuracy in time-series calculations. The analysis computes weekly increments by subtracting cumulative totals from the previous week, producing absolute and percentage changes for Russia and Ukraine separately. It also calculates the ratio of Russian to Ukrainian losses and compares these to the prior week to assess relative attrition trends. Beyond weekly differences, the module incorporates a retrospective component, analyzing up to one year of historical data to contextualize the current week's figures against long-term averages and identifying the most significant weekly spikes. The analysis extends to category-level dynamics, evaluating which equipment types recorded the highest increases or decreases relative to their historical averages, thereby identifying emerging tactical or operational trends. The results are formatted as two outputs: a structured statistical summary and a natural-language narrative enriched with retrospective context and category-level interpretation.

Module 3: Occupation Change Analysis (DeepStateMap) evaluates territorial control dynamics by comparing geospatial frontline data from DeepStateMap across a one-week interval. The module combines daily polygons of occupied areas with official oblast-level administrative boundaries, both projected into an equal-area coordinate system to enable precise areal calculations. Occupied and total oblast areas are computed through geometric intersections, producing the absolute extent of occupied territory in square kilometers as well as the percentage of each oblast under control. The analysis first establishes national-level occupation figures for the latest date and contrasts them with values from seven days earlier, calculating absolute and percentage changes to capture weekly territorial shifts. It then disaggregates results at the oblast level, ranking regions by occupation percentage and highlighting the three with the largest increases and decreases in occupied area. A significance filter is applied to identify oblasts with more than ten percent of their territory under occupation, providing a concise set of regions for targeted monitoring. The narrative consists of a short executive-style summary describing national-level occupation and change, and a more detailed report listing oblast-level increases, decreases, and significant cases.

Module 4: Settlements (DeepStateMap) extends the frontline analysis by linking geospatial control changes to populated places, providing a settlement-level perspective on the war's dynamics. The module integrates daily occupied-area polygons from DeepStateMap with point data of Ukrainian settlements (cities, towns, villages) and oblast boundaries derived from official administrative datasets. Geometries are reprojected into a metric coordinate system to ensure accurate distance and area measurements. Weekly gains and losses are computed by differencing occupied-area unions from the most recent seven-day period with those from the preceding week, yielding net Russian gains and Ukrainian recaptures in square kilometers. These figures are contextualized against the previous two weeks to calculate percentage changes in control dynamics. Newly occupied and newly liberated settlements are identified through geometric intersections of the frontline change polygons with settlement points, while oblast-level aggregation highlights the regions experiencing the most pronounced shifts in control. The analysis further estimates the average advance and retreat distances of frontline segments and applies dynamic buffers to detect settlements most at risk of being directly affected in the near term. The outputs consist of a narrative summary describing weekly territorial dynamics, settlement gains and losses, and oblast-level hotspots, supplemented by a forward-looking report listing threatened settlements based on proximity to recent frontline movement.

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Module 5: Occupation Projections (DeepStateMap) extends the static occupation analysis by introducing forward-looking estimates of territorial control. The module begins with daily DeepStateMap frontline polygons reprojected into an equal-area system and intersects them with oblast-level administrative boundaries. Occupied area and percentage are calculated for both the most recent and the preceding week, producing absolute and relative changes. These figures form the basis of simple linear projections: if an oblast's occupied area has increased, the weekly increment is extrapolated to estimate the time required for full occupation, expressed in both weeks and years. Conversely, where no gains are observed, occupation is assessed as stalled and no projection is generated. The results are organized hierarchically, beginning with national-level occupation dynamics and continuing with oblast-level projections for regions with partial occupation—defined as between two and ninety-five percent. This filtering ensures that analytical attention is focused on contested territories rather than those almost fully occupied or almost entirely liberated. Outputs consist of structured CSV tables with current, past, and projected values, accompanied by a narrative report. The narrative provides an OSINT-style briefing summarizing overall occupation trends, highlighting oblasts with significant changes, and discussing likely trajectories under current conditions. While the projections offer intuitive insights into possible future scenarios, they remain conditional, assuming the continuation of short-term trends. As with all extrapolations, the results are sensitive to fluctuations in operational tempo, sudden offensives, or withdrawals, and should therefore be treated as indicative rather than predictive.

Module 6: Maps (DeepStateMap) provides a visual representation of frontline developments by converting territorial change into geospatial figures. The module processes daily occupied-area polygons from DeepStateMap, reprojects them into a metric coordinate system, and calculates differences between consecutive days over the past week. Gains and losses are expressed as new geometries, each with associated area in square kilometers. The largest three gains and losses for each day are extracted and visualized on static maps, using red polygons for Russian advances and blue polygons for Ukrainian recaptures, overlaid on an OpenStreetMap basemap for geographic context. Each map includes an inset annotation of the areal extent of the change, while a centered title reports daily totals of territorial gains and losses. These daily map panels are structured as a chronological sequence of figures with headings summarizing the balance of territorial shifts for each day. The report provides both a high-level overview of weekly dynamics and a fine-grained spatial account of where control has changed. By integrating geospatial analysis with cartographic outputs, the module transforms numerical calculations of gains and losses into immediately interpretable visual evidence, ensuring that territorial trends are communicated in an accessible form. Limitations include the resolution of input polygons, which may generalize frontline positions, and the focus on the top three changes per side, which may omit smaller but operationally relevant shifts.

Module 7: Graphs and Pie Charts (DeepStateMap) complements the cartographic outputs with a quantitative visualization of territorial changes over the past week. The module calculates daily gains and losses by comparing consecutive occupied-area polygons, intersecting the resulting geometries with administrative oblast boundaries to attribute changes to specific regions. The outputs are aggregated into a structured dataset capturing Russian and Ukrainian area changes in square kilometers, disaggregated by date and oblast. This information is then transformed into a stacked bar chart showing the scale and composition of daily advances and retreats, with separate columns for Russian occupation and Ukrainian liberation. Distinctive colors assigned to each oblast allow for quick identification of where changes occurred, while patterned hatching differentiates Ukrainian gains from Russian gains. Complementing the time-series perspective, proportional pie charts summarize the distribution of total gains by oblast for each side, scaled in size according to overall magnitude. This dual visualization highlights both temporal dynamics and spatial concentration of activity, making it clear which regions contributed most to shifts in control. The module also generates a statistical summary of the week's developments, including total gains for each side, the three most affected oblasts, and the dates on which peak advances occurred. Limitations stem from the resolution of frontline data and the aggregation of small-scale fluctuations, which may mask micro-level tactical shifts while still capturing operational-level trends.

Module 8: Weekly Territorial Map (DeepStateMap) builds a weekly territorial overview by comparing occupied frontline geometries from the DeepStateMap dataset. The method first identifies the seven most recent days of data and establishes a baseline by selecting the last available date before this weekly window. For longer-term context, a secondary baseline is also taken from one year earlier. By applying geometric operations, the method calculates both newly occupied territories (Russian gains) and liberated areas (Ukrainian gains) over the past seven days. These areas are measured in square kilometers and ranked, allowing the identification of the three largest changes on each side. The visualization combines two layers of analysis: side panels zoom into the three most significant gains for Russia and Ukraine, while the central map shows the full Ukrainian theater. This central view highlights one-year-old occupation zones in gray, overlays recent gains in red and blue, and uses directional arrows to link the side maps with

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their precise locations. To ensure interpretability, all geometries are reprojected into equal-area coordinates for accurate measurement, and then into web-mercator projection for basemap rendering. The resulting figure situates short-term developments within the broader strategic picture, showing both the absolute scale of weekly gains and their spatial distribution across the front. A concise summary quantifies total Russian and Ukrainian gains, while the side panels and arrows guide the reader to the most dynamic hotspots of the week.

Module 9: TOP3 Territorial Maps (DeepStateMap) expands the territorial change analysis by focusing on the three largest weekly gains for each side and enriching them with predictive and contextual layers. After isolating the top three polygons of change, the method generates a three-by-three grid of maps: the first column displays current occupation zones overlaid on OpenStreetMap basemaps, the second column projects possible near-future contested areas by buffering the current gains and intersecting them with opposing control, and the third column adds a terrain perspective by draping these areas over a reprojected NASA digital elevation model. To enhance interpretability, each map includes dynamic scale bars, highlighted settlement markers, and frames that visually emphasize the change area. A narrative generator complements the maps by describing the absolute and relative scale of each gain, its share of the weekly total, and the settlements currently or potentially affected if the advance continues. This produces not only cartographic but also textual intelligence, making the output suitable for briefing contexts where both visual and written situational awareness are required.

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